

Contents

Topology of the Real Numbers	2
Monday, March 9 - Open and Closed Sets	2
Wednesday, March 11 - Topological Theorems	5
Friday, March 13 - Compact Sets	8
Monday, March 16 - The Heine-Borel Theorem	9
Wednesday, March 18 - A Pathological Set: The Cantor Set	13
Friday, March 20 - Finishing up Topology	18

Definitions and Theorems to include in Analysis Journal:

- Definition 3.2.1 (open set)
- Theorem 3.2.3
- Definition 3.2.4, Theorem 3.2.5 (limit points)
- Definition 3.2.6 (isolated point)
- Definition 3.2.7 (closed set)

Readings: None

- Consider the following “definitions” of a limit point. Are any of these formulations ‘stronger’ than others?

- (1) A point x is a **limit point** of a set A if every ϵ -neighborhood of x intersects the set A at some point other than x .
- (2) A point x is a **limit point** of a set A if every ϵ -neighborhood of x contains infinitely many points in A .
- (3) A point x is a **limit point** of a set A if $x = \lim a_n$ for some sequence (a_n) such that $a_n \in A$ and $a_n \neq x$ for all $n \in \mathbb{N}$.

Given any $a \in \mathbb{R}$ and $\epsilon > 0$, the ϵ -neighborhood of a is the set

$$V_\epsilon(a) =$$

A set $G \subseteq \mathbb{R}$ is **open** if...

- (a) Is the union of two open sets open?
 - (b) Is the intersection of two open sets open?
 - (c) Do these properties hold for infinite unions or intersections?

Theorem 3.2.3

- (i) The union of an arbitrary collection of open sets is open.
- (ii) The intersection of a finite collection of open sets is open.

Proof.

□

Limit Point

A point x is a **limit point (cluster point/accumulation point)** of a set A if...

- (1) every ϵ -neighborhood of x intersects the set A at some point other than x .
- (2) every ϵ -neighborhood of x contains infinitely many points in A .
- (3) $x = \lim a_n$ for some sequence (a_n) such that $a_n \in A$ and $a_n \neq x$ for all $n \in \mathbb{N}$.

Isolated Point

A point $a \in A$ is an **isolated point** of A if...

Closed Set

A set $F \subseteq \mathbb{R}$ is **closed** if...

2. Determine if the following sets are closed and/or open, and what their limit points are.

(a) $[0, 1]$

(e) $\mathbb{Q}$

(b) $(0, 1)$

(f) $\emptyset$

(c) $(0, 1]$

(g) $\mathbb{N}$

(d) $\mathbb{R}$

(h) $A = \left\{ \frac{1}{n} : n \in \mathbb{N} \right\}$

After Class:

Pick & Prove Problems:

★ Exercise 3.2.2

Let $A = \{(-1)^n + \frac{2}{n} : n \in \mathbb{N}\}$ and $B = \{x \in \mathbb{Q} : 0 < x < 1\}$. Answer the following questions for each set:

- (a) What are the limit points?
- (b) Is the set open? Closed? (briefly explain why or why not in each case)
- (c) Does the set contain any isolated points? If so, give an example of one and explain why it is isolated.
- (d) Find the closure of the set.

★★ Exercise 3.2.4

Let A be nonempty and bounded above so that $s = \sup A$ exists.

- (a) Prove that $s \in \overline{A}$.
- (b) Can an open set contain its supremum? Give an example or prove why this is impossible.

★★ Exercise 3.2.7

Given $A \subseteq \mathbb{R}$, let L be the set of limit points of A .

- (a) Show that the set L is closed.
- (b) Argue that if x is a limit point of $A \cup L$, then x is a limit point of A . Use this observation to furnish a proof of Theorem 3.2.12.

Understanding Analysis Sections Covered: Sections 3.3, 3.4

Definitions and Theorems to include in Analysis Journal:

- Definition 3.2.11 (closure)
- Theorem 3.2.12
- Theorem 3.2.13
- Theorem 3.2.14
- Definition 3.5.3 (nowhere-dense)

Before Class:

Readings: None

Analysis Journal Prompts:

- None

Theorem 3.2.13

A set G is open if and only if _____ is closed.

A set F is closed if and only if _____ is open.

Proof.

□

Theorem 3.2.14

- (i) The union of a _____ collection of closed sets is closed.
- (ii) The intersection of a _____ collection of closed sets is closed.

Closure

Given a set $A \subseteq \mathbb{R}$ and its set of limit points L , the **closure** of A is defined to be

Theorem 3.2.12

For any $A \subseteq \mathbb{R}$, the closure $\overline{A}$ is _____ and is...

Proof in Exercise 3.2.7.

Theorem (Topological Characterization of Density)

A set $A \subseteq \mathbb{R}$ is dense in $\mathbb{R}$ if and only if

Proof in supplemental homework exercise.

Nowhere-dense

A set $A \subseteq \mathbb{R}$ is **nowhere-dense** if

After Class:

Pick & Prove Problems:

★ Exercise 3.2.8

Assume A is an open set and B is a closed set. Determine if the following sets are definitely open, definitely closed, or neither. (no justification needed)

- (a) $\overline{A \cup B}$
- (b) $A \setminus B = \{x \in A : x \notin B\}$
- (c) $(A^c \cup B)^c$
- (d) $(A \cap B) \cup (A^c \cap B)$
- (e) $\overline{A}^c \cap \overline{A}^c$

★★ Supplemental Exercise

Recall that a set A is dense in $\mathbb{R}$ if for all $x \neq y \in \mathbb{R}$, there exists some $a \in A$ such that $x < a < y$. Prove that A is dense in $\mathbb{R}$ if and only if $\overline{A} = \mathbb{R}$. (See Example 3.2.9iii for discussion).

★★★ Exercise 3.2.14

A dual notion to the closure of a set is the interior of a set. The **interior** of E is defined as

$$E^\circ = \{x \in E : \text{there exists } V_\varepsilon(x) \subseteq E \text{ for some } \varepsilon > 0\}$$

Results about closures and interiors possess a useful symmetry.

- (a) Prove that E is closed if and only if $\overline{E} = E$. Prove that E is open if and only if $E^\circ = E$.
- (b) Prove that $\overline{E}^c = (E^\circ)^\circ$, and similarly that $(E^\circ)^c = \overline{E}^c$.

Understanding Analysis Sections Covered: Sections 3.3

Definitions and Theorems to include in Analysis Journal:

- Definition 3.3.1 (sequential compactness)

Before Class:

Readings: None

Analysis Journal Prompts:

- (Part of Exercise 3.3.6) Which of the following are true when the blank is filled in with the word “finite?” Which are true if it is filled in with the word “closed?”
 1. Every _____ set has a maximum.
 2. If $\{A_n\}$ is a collection of _____ sets with the property that every finite subcollection has a nonempty intersection, then $\bigcap_{n=1}^{\infty} A_n$ is nonempty as well (Nested “interval” property).

Compact Set

A set $K \subseteq \mathbb{R}$ is **compact** if...

Compactness is an extension of _____.

1. What kinds of sets does the Bolzano-Weierstrass Theorem prove are compact? (Proof for HW)

After Class:

Pick & Prove Problems:

★★ Exercise 3.3.1

Show that if K is compact and nonempty, then $\sup K$ and $\inf K$ both exist and are elements of K . Explain why this means that every compact set must have a maximum and minimum.

★★ Exercise 3.3.3

Prove that if a set is closed and bounded, then it is compact.

Understanding Analysis Sections Covered: Section 3.3

Definitions and Theorems to include in Analysis Journal:

- Theorem 3.3.5 (Nested Compact Set Property)
- Theorem 3.3.8 (Heine-Borel Theorem)
- Definition 3.3.6 (open cover/finite subcover)

Before Class:

Readings: None

Analysis Journal Prompts:

- None

Theorem 3.3.5 (Nested Compact Set Property)

Scratch work:

Proof.

Theorem 3.3.8 (Heine-Borel Theorem)

Let K be a subset of $\mathbb{R}$. The following statements are equivalent:

(1) K is compact.

(2)

(3)

Scratch work:

(2 $\Rightarrow$ 1) Exercise 3.3.3

(1 $\Rightarrow$ 2) *Proof sketch:*

Open Cover/Finite Subcover

An **open cover** of a set $A \subseteq \mathbb{R}$ is

An **finite subcover** of an open cover is

1. $(0, 1)$ is not sequentially compact (why?). Find an open cover with no finite subcover.

More scratch work:

(2 $\Rightarrow$ 3)

(3 $\Rightarrow$ 2) Exercise 3.3.9

Proof.

□

After Class:

Pick & Prove Problems:

★ Exercise 3.3.2, 3.3.11

For the following sets, determine whether or not they are compact. If they are not compact, find an open cover for which there is no finite subcover.

- (a) $\mathbb{N}$
- (b) $\mathbb{Q} \cap [0, 1]$
- (c) $\{1, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots\}$

★ Exercise 3.3.5abc

Decide whether the following propositions are true or false. If the claim is valid, supply a brief justification, and if the claim is false, supply a counterexample.

- (a) The arbitrary intersection of compact sets is compact.
- (b) The arbitrary union of compact sets is compact.
- (c) Let A be any set, and let K be compact. Then $A \cap K$ is compact.

★★ Exercise 3.3.12

Using the concept of open covers (and explicitly avoiding the Bolzano-Weierstrass Theorem), prove that every bounded infinite set has a limit point. (You may use the definition of a limit point and the fact that every closed and bounded set is compact.)

★★★ Exercise 3.3.9 (2 stars if we have time to go over it in class)

Follow these steps to prove that any open cover of a compact set must have a finite subcover.

Assume K is compact and let there be an arbitrary open cover. For contradiction assume that no finite subcover exists. Let $K \subseteq I_0$.

- (a) Show that there exists nested sequence of closed intervals $I_0 \supseteq I_1 \supseteq I_2 \supseteq I_3 \supseteq \dots$ with the property that for each $n \in \mathbb{N}$, $I_n \cap K$ cannot be finitely covered and $\lim |I_n| = 0$.
- (b) Argue that there exists $x \in K$ such that $x \in I_n$ for all n .
- (c) Because $x \in K$, there exists an open set from the original collection that contains x as an element.

Notes: The proof in part (a) is very similar to the construction used when proving the Bolzano-Weierstrass theorem. Make sure you use both that the set is closed and bounded in your proof.

Understanding Analysis Sections Covered: Section 3.1

Definitions and Theorems to include in Analysis Journal:

- Definition 3.4.4 (separated/connected sets)

Before Class:

Readings: None

Analysis Journal Prompts:

- **Definition:** Two nonempty sets $A, B \subseteq \mathbb{R}$ are **separated** if $\overline{A} \cap B = \emptyset$ and $A \cap \overline{B} = \emptyset$.
Are the following sets separated?
 - $(0, 1)$ and $(1, 2)$
 - $(0, 1)$ and $[1, 2]$
 - The rational numbers ($\mathbb{Q}$) and the irrational numbers ($\mathbb{I}$)
- **Definition:** A set $E \subseteq \mathbb{R}$ is **disconnected** if there exist disjoint sets A and B such that $E = A \cup B$ and A and B are disconnected. A set $E \subseteq \mathbb{R}$ is **connected** if it is not disconnected.
Are the following sets connected or disconnected?
 - $\{\frac{1}{n} : n \in \mathbb{N}\}$
 - The rational numbers ($\mathbb{Q}$)

Constructing The Cantor “Middle Thirds” Set

1. What is left? What is in this infinite intersection?

The Cantor Set has *zero length*.

The **topological (Hausdorff) dimension** of the Cantor Set is _____.

Questions to answer:

1. Is C countable or uncountable?
2. Is C open?
3. Is C closed?
4. Are there any isolated points in C ?
5. Is C compact?
6. Is C connected?

After Class:

Pick & Prove Problems:

★ Exercise 3.4.4

Repeat the Cantor construction starting with the interval $[0,1]$, but this time remove the middle half from each interval. (“Middle half” is a little oxymoronic, we mean removing the middle 50%. So our second set would be $[0, 1/4] \cup [3/4, 1]$.)

Briefly answer the following questions about how this new set compares to the “middle thirds” Cantor set.

- (a) Does the resulting set still have no length? (Do a sum to verify)
- (b) Is the resulting set also uncountable?
- (c) What is the dimension of this new set? (Use a similar method to what we did in class to calculate this)
- (d) Is this new set open? Closed? Compact? Connected?

The upshot of this exercise is that we can create similar “Cantor Dusts” of different dimensions by removing different portions of each included interval.

★★ Exercise 3.4.7

A set E is **totally disconnected** if, given any two distinct points $x, y \in E$, there exist separated sets A and B with $x \in A, y \in B$ and $E = A \cup B$.

- (a) Prove that the set of rational numbers is totally disconnected.
- (b) Is the set of irrational numbers totally disconnected?

★★ Exercise 3.4.8

Follow these steps show that the Cantor set is totally disconnected.

- (a) Given $x, y \in C$ with $x < y$, set $\epsilon = y - x$. For each $n \in \mathbb{N}$, the set C_n consists of a finite number of closed intervals. Explain why there must exist an N large enough so that it is impossible that x and y both to belong to the same closed interval of C_N .
- (b) Show that the C is totally disconnected.

★★ Supplemental Exercise

Prove that the Cantor set has no isolated points.

★ ★ ★ Exercise 3.3.7

Follow the exercise in the book to prove another surprising fact about the Cantor Set: The sum $C + C = \{x + y : x, y \in C\}$ is equal to the closed interval $[0, 2]$. (it’s too long for me to type out here)

Understanding Analysis Sections Covered: Sections 3.4

Definitions and Theorems to include in Analysis Journal:

- Definition of an interval

Before Class:

Readings: None

Analysis Journal Prompts:

- We've know about several types of intervals: (a, b) , $[a, b]$, (a, ∞) , etc. How would you write a definition of an interval?

Theorem 3.4.7

A set is connected if and only if it is an _____.

Definition of an Interval

A set $I \subseteq \mathbb{R}$ is an **interval** if I has at least two points and...

1. We know a *closed set* is a generalization of the idea of a *closed interval*. Is every closed set a union of disjoint closed intervals?
2. We know a *compact set* is also a generalization of the idea of a closed interval. Is every compact set a union of disjoint closed intervals?
3. We know a *open set* is a generalization of the idea of an *open interval*. Is every open set a union of disjoint open intervals?

After Class:

Pick & Prove Problems:

★★ Supplemental Exercise

Follow the steps to write a proof that every open set G is the union of countably many disjoint open intervals (of the form (a, b) , $(-\infty, b)$, or (a, ∞)).

- (a) For each $x \in G$, let $I_x \subseteq G$ be the largest open interval containing x . Argue that for any $x, y \in G$, either $I_x = I_y$ or $I_x \cap I_y = \emptyset$.
- (b) Show that $\bigcup_{x \in A} I_x = G$, and that there are only countably many distinct intervals.